Introduction to HTML

I. What is HTML?

Let us begin with a simple question: *How does a webpage actually exist?* When we open a website, what we see—text, images, buttons—is not magic. It is structured using a language called **HTML**, which stands for **HyperText Markup Language**.

HTML is not a programming language in the traditional sense. It does not perform logic or calculations. Instead, we use it to *structure* content. Think of HTML as the skeleton of a human body—it provides the framework upon which everything else is built.

We use HTML to tell the browser:

- "This is a heading."
- "This is a paragraph."
- "This is an image."
- "This is a link."

Without HTML, a webpage would be like a book with no chapters, no paragraphs, and no organization—just chaos.

II. Basic Structure of an HTML Document

Now, let us peel back the layers. Every HTML document follows a standard structure. It is like a template we must respect.

1. The Core Components

An HTML document typically looks like this:

```
<!DOCTYPE html>
<html>
<head>
<title>My First Page</title>
</head>
<body>
<h1>Hello World!</h1>
<p>This is my first webpage.</p>
</body>
</html>
```

Let us break this down together.

- `<!DOCTYPE html>` : This tells the browser that we are using HTML5.
- `<html>` : This is the root element—everything lives inside it.
- `<head>` : Contains meta-information like the page title.
- `<body>` : This is where visible content goes.

2. Why Structure Matters

Imagine building a house without a blueprint. Sounds risky, right? HTML structure acts as our blueprint. If we misuse it, the browser may still try to render the page—but the result might be messy or unpredictable.

III. HTML Elements and Tags

1. Understanding Tags

HTML uses **tags** to define elements. Most tags come in pairs:

```
<p>This is a paragraph</p>
```

Here:

- `<p>` is the opening tag
- `</p>` is the closing tag

Between them lies the content.

2. Common HTML Elements

Let us explore a few essential ones:

- **Headings**:

```
<h1>Main Heading</h1>
<h2>Subheading</h2>
```

- **Paragraphs**:

```
<p>This is a paragraph.</p>
```

- **Links**:

```
<a href="https://example.com">Visit Site</a>
```

- **Images**:

```
<img src="image.jpg" alt="Description">
```

Notice something interesting? The `<img>` tag does not need a closing tag. This is called a **self-closing tag**.

3. Nesting Elements

HTML elements can be placed inside one another:

```
<p>This is <strong>important</strong> text.</p>
```

We call this **nesting**. It is like placing boxes inside boxes—organization within organization.

IV. Attributes and Their Role

1. What Are Attributes?

Attributes provide additional information about elements. They are placed inside the opening tag.
Example:

```
<a href="https://example.com">Click here</a>
```

Here, `href` is an attribute that defines the link destination.

2. Common Attributes

- `href` : Used in links
- `src` : Used in images
- `alt` : Describes images (important for accessibility)
- `class` and `id` : Used for styling and scripting

3. Why Attributes Matter

Think of attributes as adjectives in a sentence. If HTML elements are nouns, attributes describe them. Without attributes, our webpage would lack detail and functionality.

V. Semantic HTML and Best Practices

1. What is Semantic HTML?

Semantic HTML means using tags that clearly describe their purpose.
For example:

- `<header>` instead of a generic `<div>`
- `<article>` for content
- `<footer>` for the bottom section

Why does this matter? Because it improves:

- Readability
- Accessibility
- Search engine optimization (SEO)

2. Clean and Readable Code

We should always aim for clarity. Consider this:
Bad practice:

```
<div><div><p>Text</p></div></div>
```

Better:

```
<article>
<p>Text</p>
</article>
```

We must not forget users with disabilities. Using proper tags and attributes (like `alt` for images) ensures everyone can access our content. Ask yourself: *Can someone using a screen reader understand my page?* If the answer is yes, we are on the right track.

VI. HTML in the Bigger Picture

HTML does not work alone. It is part of a trio:

- **HTML** → Structure
- **CSS** → Style
- **JavaScript** → Behavior

Think of it like this:

- HTML is the **skeleton**
- CSS is the **clothing**
- JavaScript is the **movement**

Without HTML, the other two would have nothing to act upon.

Conclusion

As we conclude, let us reflect on what we have explored. HTML is the foundation of every webpage. It gives structure, meaning, and organization to content. We learned about its basic structure, elements, attributes, and best practices. More importantly, we understood *why* each part matters.

So, the next time you open a webpage, ask yourself: *What does its HTML look like beneath the surface?* That curiosity is the first step toward mastering web development.

Remember, mastering HTML is not about memorizing tags—it is about understanding structure. Once we grasp that, everything else becomes easier.